



# Two Classes of Quantum Synchronizable Codes

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Received: 10 April 2022 / Accepted: 14 June 2022 / Published online: 15 October 2022

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## Abstract

Quantum synchronizable codes(QSCs) are special classes of quantum error-correcting codes that can correct not only the effects of quantum noise on qubits but also the misalignment in block synchronization. In this paper, we construct two classes of QSCs from cyclic codes. The first class is to construct cyclic and negacyclic codes of odd prime length by using cyclotomic polynomials, and obtain a class of QSCs which achieves the best attainable misalignment under certain conditions. The second class is to apply BCH codes and negacyclic BCH codes on  $(u + v)|(u - v)$  structure to construct a class of QSCs.

**Keywords** QSCs · Cyclotomic polynomials · Cyclic codes · BCH codes · Negacyclic BCH code

## 1 Introduction

In 1995, Shor proposed the first method to construct quantum code. In 1996, Calderbank and Shor and Steane proposed the famous Calderbank-Shor-Steane (CSS) construction. It provides an effective method for constructing quantum codes. In recent years, quantum error-correcting codes have received extensive attention and research, and many research branches have been produced such as asymmetric quantum codes, entanglement-assisted quantum codes, quantum synchronization codes and so on.

Fujiwara proposed the concept of QSCs and gave a construction method in 2013, and further improved the method of constructing QSC in [1], which the  $c_r + c_l$  in QSCs can reach the order of quotient of two generator polynomials. Fujiwara et al. [5] constructed two classes of QSCs by using narrow-sense Bose-Chaudhuri-Hocquenghem (BCH) codes and punctured Reed-Muller codes. The construction of QSCs has received much attention during these years. In 2018, Luo and Ma [12] constructed a new family of QSCs from repeated-root cyclic codes of lengths  $p^s$  and  $lp^s$  over  $F_q$ , where  $s \geq 1$  and  $l \geq 2$  are integers. In 2021, Liu et al. [16] provided two methods of constructing QSCs

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from cyclic or constacyclic codes over finite rings. Xia et al. [10] constructed dual-containing cyclic codes from cyclotomic polynomials, and gave a class of QSCs.

The  $(u + v)|(u - v)$  construction has many good properties over the field. Hughes [7] constructed an optimal cyclic code of length  $2(q + 1)$  over  $F_q$  by using  $(u + v)|(u - v)$  construction. Luo and Ma et al. [12] constructed QSCs with good parameters by using  $(u + v)|(u - v)$  construction. Du et al. [15] constructed QSCs based on  $\lambda(u + v)|(u - v)$  construction by using constacyclic codes and repeated-root cyclic codes.

In this paper, we construct two classes of QSCs by using  $(u + v)|(u - v)$  construction. We apply cyclic codes and negacyclic codes on  $(u + v)|(u - v)$  construction to obtain QSCs, which can achieve the best attainable misalignment tolerance under certain constraints.

## 2 Preliminaries

Let  $F_q$  be a finite field of characteristic  $p$  where  $q = p^s, s \geq 1$ . A linear code of length  $n$  over  $F_q$  is a subspace of  $F_q^n$ . A linear code  $C$  of length  $n$  is called a  $\lambda$ -constacyclic code if  $(\lambda c_{n-1}, c_0, c_1, c_2, \dots, c_{n-2}) \in C$  for  $\forall (c_0, c_1, \dots, c_{n-1}) \in C$ . By identifying a vector  $(c_0, c_1, c_2, \dots, c_{n-1}) \in F_q^n$  with

$$c_0 + c_1x + c_2x^2 + \dots + c_{n-1}x^{n-1} \in F_q[x]/\langle x^n - \lambda \rangle,$$

it is known that a linear code  $C$  can be regarded as an ideal of the principal ring  $F_q[x]/\langle x^n - \lambda \rangle$  if and only if  $C$  is a  $\lambda$ -constacyclic code. If  $\lambda = 1$ ,  $C$  is just a cyclic code, and if  $\lambda = -1$ ,  $C$  is a negacyclic code.

For a  $\lambda$ -constacyclic code  $C$ , there exists a monic divisor of  $x^n - \lambda$  such that  $C = \langle g(x) \rangle$ . The polynomial  $g(x)$  is called the generator polynomial of  $C$ , and  $h(x) = \frac{x^n - \lambda}{g(x)}$  is called the parity check polynomial of  $C$ . The dual code of  $C$  is defined as

$$C^\perp = \left\{ x \in F_q^n \mid (x, y) = 0, \forall y \in C \right\},$$

where  $(x, y)$  denotes the inner product of  $x$  and  $y$ . If  $C^\perp \subseteq C$ , then  $C$  is called a dual-containing  $\lambda$ -constacyclic code. If  $\lambda = 1$ , then  $C$  is a cyclic code. Moreover  $C^\perp$  is also a cyclic code, and its generator polynomial is

$$h^*(x) = x^{\deg(h(x))} h(x^{-1}),$$

where  $h^*(x)$  is reciprocal polynomial of  $h(x)$ .

Let  $\gcd(n, q) = 1$ , and  $q$  be an odd prime power and we assume that  $m = \text{ord}_{2n}(q)$ . Let  $\xi$  be a  $2n$ -th primitive root of unity in  $F_{q^m}$ . Then the root of  $(x^n - 1)$  are  $\xi^{2i}, 0 \leq i \leq n - 1$ , and the root of  $(x^n + 1)$  are  $\xi^{1+2j}, 0 \leq j \leq n - 1$ .

The polynomial

$$Q_{2n}(x) = \prod_{s=0, (s, 2n)=1}^{2n} (x - \xi^s)$$

is called the  $2n$ -th cyclotomic polynomial over  $F_q$ , and  $x^{2n} - 1 = \prod_{d|2n} Q_d(x)$ .

Let  $\eta = \xi^2$  be an  $n$ -th primitive root of unity over  $F_q$ , the polynomial

$$Q_n(x) = \prod_{s=0, (s,n)=1}^n (x - \eta^s)$$

is called the  $n$ -th cyclotomic polynomial over  $F_q$ , and  $x^n - 1 = \prod_{d|n} Q_d(x)$ .

Let

$$\Omega_1 = \{1 + 2j : 0 \leq j \leq n-1\},$$

$$\Omega_2 = \{2j : 0 \leq j \leq n-1\},$$

and  $\Lambda$  be the set of all the coset leaders modulo  $2n$ .

Let

$$\Lambda_1 = \{i \in \Lambda : i \text{ is odd}\} \text{ and } \Lambda_2 = \{i \in \Lambda : i \text{ is even}\}.$$

For a nonnegative integer  $i \in \Lambda$ , let  $C_i^{(2n)} = \{i, iq, iq^2, \dots, iq^{l_i-1}\}$ , where  $l_i$  is the smallest positive integer such that  $iq^{l_i} \equiv i \pmod{2n}$ . The set  $C_i^{(2n)}$  is called the  $q$ -cyclotomic coset modulo  $2n$  containing  $i$ . Then  $\Omega_1 = \bigcup_{i \in \Lambda_1} C_i^{(2n)}$  and  $\Omega_2 = \bigcup_{i \in \Lambda_2} C_i^{(2n)}$ .

Let  $m_i^{(2n)}(x)$  be the minimum polynomial of  $\xi^i$  over  $F_q$ , then

$$x^n + 1 = \prod_{i \in \Lambda_1} m_i^{(2n)}(x) \text{ and } x^n - 1 = \prod_{i \in \Lambda_2} m_i^{(2n)}(x)$$

where  $m_i^{(2n)}(x) = \prod_{j \in C_i^{(2n)}} (x - \xi^j)$ .

Let  $C_1$  and  $C_2$  be cyclic and negacyclic code of length  $n$  with generator polynomial  $g_1(x)$  and  $g_1(x)$  respectively. The defining set of  $C_1$  is defined as the set  $Z_1 = \{i \in \Omega_1 : g_1(\xi^i) = 0\}$ , and the defining set of  $C_2$  is defined as the set  $Z_2 = \{j \in \Omega_2 : g_2(\xi^j) = 0\}$ .

### 3 The Constructions of QSCs

Let  $C_i^n = \{i, iq, iq^2, \dots, iq^{l_i-1}\}$  denote the  $q$ -cyclotomic coset modulo  $n$  containing  $i$ . Define  $T_0 = \{d : q^{l_d} \equiv -1 \pmod{n}\}$ ,  $T_1 = \{d : q^{l_d} \not\equiv -1 \pmod{n}\}$  where  $1 \leq l_d \leq l_1 - 1$ . Then  $x^n - 1 = \prod_{d|n} Q_d(x) = \prod_{d \in T_0} Q_d(x) \prod_{d \in T_1} Q_d(x)$ .

**Lemma 3.1** [10] Let  $\gcd(n, q) = 1$ ,  $\phi(d)/2t_d = \omega$ . Then

$$\prod_{d \in T_1} Q_d(x) = \prod_{d \in T_1} f_{d,1}(x) f_{d,1}^*(x) \cdots f_{d,\omega}(x) f_{d,\omega}^*(x), \quad (3.1)$$

Where  $f_{d,j}^*(x), f_{d,j}(x)$  is a reciprocal polynomial pair,  $f_{d,j}^*(x) \neq f_{d,j}(x)$  and  $\deg(f_{d,j}(x)) = t_d, 1 \leq j \leq \omega$ .

**Theorem 3.2** Let notations as in Lemma 3.1. Let  $C_1 = \langle g_1(x) \rangle$  be an  $[n, k_1, d_1]$  cyclic code over  $F_q$  and  $C_3 = \langle g_3(x) \rangle$  be an  $[n, k_3, d_3]$  cyclic code over  $F_q$ , where

$$g_1(x) = \prod_{d \in T_1} f_{d,1}^{\varepsilon_{d,1}}(x) f_{d,2}^{\varepsilon_{d,2}}(x) \cdots f_{d,\omega}^{\varepsilon_{d,\omega}}(x), \quad (3.2)$$

$$g_3(x) = \prod_{d \in T_1} f_{d,1}^{o_{d,1}}(x) f_{d,2}^{o_{d,2}}(x) \cdots f_{d,\omega}^{o_{d,\omega}}(x), \quad (3.3)$$

each  $\varepsilon_{d,j}, o_{d,j} \in \{0, 1\}$ ,  $\varepsilon_{d,j} \geq o_{d,j}$ ,  $1 \leq j \leq \omega$ . If  $\sum_{d \in T_1} \sum_{j=1}^{\omega} \varepsilon_{d,j} > \sum_{d \in T_1} \sum_{j=1}^{\omega} o_{d,j} \geq 1$ , then  $C_1^\perp \subset C_1$  and  $C_1 \subset C_3$ .

**Proof** Note that  $\varepsilon_{d,j} \geq o_{d,j}$  and  $\sum_{d \in T_1} \sum_{j=1}^{\omega} \varepsilon_{d,j} > \sum_{d \in T_1} \sum_{j=1}^{\omega} o_{d,j} \geq 1$ . Since (3.2), (3.3) are generator polynomials of  $C_1$  and  $C_3$ , respectively, then  $g_3(x)|g_1(x)$ , i.e.,  $C_1 \subset C_3$ . Since

$$g_1(x) = \prod_{d \in T_1} f_{d,1}^{\varepsilon_{d,1}}(x) f_{d,2}^{\varepsilon_{d,2}}(x) \cdots f_{d,\omega}^{\varepsilon_{d,\omega}}(x),$$

$f_{d,j}^*(x) \neq f_{d,j}(x)$ , then by Lemma 3.1  $C_1^\perp \subset C_1$ .

Since  $Q_{2n}(x) = \prod_{s=0, (s, 2n)=1}^{2n} (x - \xi^s)$ , where  $s$  is odd, then  $Q_{2n}(x)|x^n + 1$ . Let  $x^n + 1 = N(x)Q_{2n}(x)$ . If  $n$  is an odd prime,  $\deg(Q_{2n}(x)) = \phi(2n) = n - 1$ . And  $-1$  is not a root of  $Q_{2n}(x)$ , we can obtain that  $x^n + 1 = (x + 1)Q_{2n}(x)$ .  $\square$

**Theorem 3.3** Let  $n$  be an odd prime,  $\gcd(n, q) = 1$  and  $-1 \notin C_1^{2n}$ . Let  $\kappa = \frac{(n-1)}{2|C_1^{(2n)}|}$ , then

$$Q_{2n}(x) = y_1(x)y_1^*(x) \cdots y_\kappa(x)y_\kappa^*(x),$$

where  $y_j^*(x), y_j(x)$  is a reciprocal polynomial pair, and  $y_j^*(x) \neq y_j(x)$ ,  $1 \leq j \leq \kappa$ .

**Proof** The minimal polynomial  $m_1^{(2n)}(x)$  of  $\xi$  over  $F_q$  is

$$m_1^{(2n)}(x) = \prod_{j \in C_1^{(2n)}} (x - \xi^j).$$

Since  $Q_{2n}(x) = \prod_{s=0, (s, 2n)=1}^{2n} (x - \xi^s)$ , then  $m_1^{(2n)}(x)|Q_{2n}(x)$ . Note that  $\deg(m_1^{(2n)}(x)) = |C_1^{2n}|$ , so  $Q_{2n}(x)$  can factorize into  $(n-1)/|C_1^{2n}|$  distinct monic irreducible polynomials of degree  $|C_1^{2n}|$  in  $F_q$ .

Note that  $-1 \notin C_1^{2n}$ , it is clear that  $q^{l_{2n}} \not\equiv -1 \pmod{2n}$ , then

$$Q_{2n}(x) = y_1(x)y_1^*(x) \cdots y_\kappa(x)y_\kappa^*(x),$$

where  $y_j^*(x) \neq y_j(x)$ , for  $1 \leq j \leq \kappa$  and  $y_j^*(x)$  is a reciprocal polynomial of an irreducible polynomial  $y_j(x)$ .  $\square$

**Theorem 3.4** Let  $n$  be an odd prime and  $\kappa = \frac{(n-1)}{2|C_1^{(2n)}|}$ . Let  $C_2 = \langle g_2(x) \rangle$  be a  $[n, k_2, d_2]$  negacyclic code over  $F_q$  and  $C_4 = \langle g_4(x) \rangle$  be a  $[n, k_4, d_4]$  negacyclic code over  $F_q$ , where

$$g_2(x) = y_1^{u_1}(x)y_2^{u_2}(x) \cdots y_\kappa^{u_\kappa}(x), \quad (3.4)$$

$$g_4(x) = y_1^{v_1}(x)y_2^{v_2}(x)\cdots y_\kappa^{v_\kappa}(x), \quad (3.5)$$

each  $u_j, v_j \in \{0, 1\}$ ,  $u_j \geq v_j$ ,  $1 \leq j \leq \kappa$ . If  $\sum_{j=1}^{\kappa} u_j > \sum_{j=1}^{\kappa} v_j \geq 1$ , then  $C_2^\perp \subset C_2$  and  $C_2 \subset C_4$ .

**Proof** Since  $u_j, v_j \in \{0, 1\}$  and  $\sum_{j=1}^{\kappa} u_j > \sum_{j=1}^{\kappa} v_j \geq 1$ , then  $g_4(x)|g_2(x)$ , i.e.,  $C_2 \subset C_4$ .

Since  $g_2(x) = y_1^{u_1}(x)y_2^{u_2}(x)\cdots y_\kappa^{u_\kappa}(x)$ , and  $g_2(x)|Q_{2n}(x)|(x^n + 1)$ , then the generator polynomial of  $C_2^\perp$  is defined as

$$h_2^*(x) = \left( \frac{(x+1)Q_{2n}(x)}{g_2(x)} \right)^* = \frac{(x+1)y_1(x)y_1^*(x)\cdots y_\kappa(x)y_\kappa^*(x)}{y_1^{u_1}(x)y_2^{u_2}(x)\cdots y_\kappa^{u_\kappa}(x)},$$

where  $u_j \in \{0, 1\}$ , then  $g_2(x)|h_2^*(x)$ , i.e.,  $C_2^\perp \subset C_2$ .  $\square$

In [7], we assume that  $C_1$  and  $C_2$  be a  $[n, k_1, d_1]$  and  $[n, k_2, d_2]$  linear codes of length  $n$  over  $F_q$ , respectively, then  $(u+v)|(u-v)$  construction  $C_1 \vee C_2 = \{(u+v, u-v) : u \in C_1, v \in C_2\}$  is a  $[2n, k_1+k_2, \min\{2d_1, 2d_2, \max\{d_1, d_2\}\}]$ .  $C_1 \vee C_2$  is a linear code of length  $2n$  over  $F_q$ . Specifically,  $C_1$  be a  $[n, k_1, d_1]$  cyclic code of length  $n$  and  $C_2$  be a  $[n, k_2, d_2]$  negacyclic code of length  $n$  over  $F_q$ , then  $C_1 \vee C_2$  be a cyclic code of length  $2n$  over  $F_q$ . We can obtain that  $C_1 \vee C_2$  be a  $[2n, k_1+k_2, d]$  cyclic code with  $d = \min\{2d_1, 2d_2, \max\{d_1, d_2\}\}$ , and  $C_3 \vee C_4$  be a  $[2n, k_3+k_4, d^*]$  cyclic code with  $d^* = \min\{2d_3, 2d_4, \max\{d_3, d_4\}\}$ .  $C_1 \vee C_2$  is dual-containing if  $C_1$  and  $C_2$  are dual-containing respectively.

The following theorem is an important construction of QSCs given by Luo et al. [7].

**Theorem 3.5** Let  $n$  be an odd prime, and  $C_i$  be an  $[n, k_i, d_i]$  dual-containing code over  $F_q$  for  $i=1,2,3,4$ . Suppose that  $C_1, C_3$  are cyclic codes with respective generator polynomial  $g_1(x), g_3(x)$  and  $C_2, C_4$  are negacyclic with respective generator polynomial  $g_2(x), g_4(x)$ . If  $C_1 \subset C_3$  and  $C_2 \subset C_4$ , define  $f(x) = \frac{g_1(x)g_3(x)}{g_2(x)g_4(x)}$ . Then for any pair of non-negative integers  $a_l, a_r$  such that  $(a_l + a_r) < \text{ord}(f(x))$ , there exists an  $(a_l, a_r) - [[2n + a_l + a_r, 2(k_1 + k_2) - 2n]]$  QSC that can correct up to  $\left\lfloor \frac{\min\{2d_3, 2d_4, \max\{d_3, d_4\}\} - 1}{2} \right\rfloor$  bit errors and  $\left\lfloor \frac{\min\{2d_1, 2d_2, \max\{d_1, d_2\}\} - 1}{2} \right\rfloor$  phase errors.

Now we can obtain a family of QSC.

**Theorem 3.6** Let  $n$  be an odd prime. Let  $C_i = \langle g_i(x) \rangle$ ,  $i=1,2,3,4$  be  $[n, k_i, d_i]$  linear codes of length  $n$  over  $F_q$ . Suppose that  $C_1 = \langle g_1(x) \rangle$  and  $C_3 = \langle g_3(x) \rangle$  be cyclic codes, and  $C_2 = \langle g_2(x) \rangle$  and  $C_4 = \langle g_4(x) \rangle$  be negacyclic codes. Where

$$g_1(x) = \prod_{d \in T_1} f_{d,1}^{\varepsilon_{d,1}}(x) f_{d,2}^{\varepsilon_{d,2}}(x) \cdots f_{d,\omega}^{\varepsilon_{d,\omega}}(x), \quad (3.6)$$

$$g_2(x) = y_1^{u_1}(x)y_2^{u_2}(x)\cdots y_\kappa^{u_\kappa}(x), \quad (3.7)$$

$$g_3(x) = \prod_{d \in T_1} f_{d,1}^{\sigma_{d,1}}(x) f_{d,2}^{\sigma_{d,2}}(x) \cdots f_{d,\omega}^{\sigma_{d,\omega}}(x), \quad (3.8)$$

$$g_4(x) = y_1^{v_1}(x)y_2^{v_2}(x)\cdots y_k^{v_k}(x). \quad (3.9)$$

Let

$$f(x) = \prod_{d \in T_1} f_{d,1}^{\varepsilon_{d,1}-o_{d,1}}(x) f_{d,2}^{\varepsilon_{d,2}-o_{d,2}}(x) \cdots f_{d,\omega}^{\varepsilon_{d,\omega}-o_{d,\omega}}(x) y_1^{u_1-v_1}(x) y_2^{u_2-v_2}(x) \cdots y_k^{u_k-v_k}(x).$$

Then for a pair of non-negative integers  $(a_l, a_r)$ ,  $(a_l + a_r) < \text{ord}(f(x))$ , there exists an  $(a_l, a_r) - \lfloor [2n + a_l + a_r, 2(k_1 + k_2) - 2n] \rfloor$  QSC that can correct up to  $\left\lfloor \frac{d^* - 1}{2} \right\rfloor$  bit errors and  $\left\lfloor \frac{d - 1}{2} \right\rfloor$  phase errors, where  $d = \min\{2d_1, 2d_2, \max\{d_1, d_2\}\}$  and  $d^* = \min\{2d_3, 2d_4, \max\{d_3, d_4\}\}$ .

**Proof** According to the conditions, we can obtain that  $C_1 \vee C_2$  is a  $[2n, k_1 + k_2, d]$  cyclic code with  $d = \min\{2d_1, 2d_2, \max\{d_1, d_2\}\}$ , and  $C_3 \vee C_4$  is a  $[2n, k_3 + k_4, d^*]$  cyclic code with  $d^* = \min\{2d_3, 2d_4, \max\{d_3, d_4\}\}$ . Since generator polynomials  $g_1(x)$ ,  $g_2(x)$ ,  $g_3(x)$  and  $g_4(x)$  satisfy Theorems 3.2 and 3.4, it is clear that  $C_1^\perp \vee C_2^\perp \subset C_1 \vee C_2 \subset C_3 \vee C_4$ . Defined

$$f(x) = \frac{g_1(x)g_3(x)}{g_2(x)g_4(x)} = \prod_{d \in T_1} f_{d,1}^{\varepsilon_{d,1}-o_{d,1}}(x) f_{d,2}^{\varepsilon_{d,2}-o_{d,2}}(x) \cdots f_{d,\omega}^{\varepsilon_{d,\omega}-o_{d,\omega}}(x) y_1^{u_1-v_1}(x) y_2^{u_2-v_2}(x) \cdots y_k^{u_k-v_k}(x).$$

By Theorem 3.5, for a pair of non-negative integers  $(a_l, a_r)$ ,  $(a_l + a_r) < \text{ord}(f(x))$ , there exists an  $(a_l, a_r) - \lfloor [2n + a_l + a_r, 2(k_1 + k_2) - 2n] \rfloor$  QSC.  $\square$

To approach the highest tolerance against misalignment, the pair of cyclic codes and negacyclic codes should be chosen such that  $\text{ord}(f(x))$  is as large as possible.

**Theorem 3.7** Let the notations be as Theorem 3.6. Let

$$\sum_{j=1}^{\omega} \varepsilon_{d,j} > \sum_{j=1}^{\omega} o_{d,j}, \sum_{j=1}^K u_j > \sum_{j=1}^K v_j.$$

Then for a pair of non-negative integers  $(a_l, a_r)$ ,  $(a_l + a_r) < 2n$ , there exists an  $(a_l, a_r) - \lfloor [2n + a_l + a_r, 2(k_1 + k_2) - 2n] \rfloor$  QSC.

**Proof** Since  $\sum_{j=1}^{\omega} \varepsilon_{d,j} > \sum_{j=1}^{\omega} o_{d,j}$  and  $\sum_{j=1}^K u_j > \sum_{j=1}^K v_j$ , then there are irreducible factors of  $Q_n(x)$  such that  $Q_{2n}(x)$  in  $f(x)$  and  $\text{ord}(f(x)) = 2n$ . We can obtain an  $(a_l, a_r) - \lfloor [2n + a_l + a_r, 2(k_1 + k_2) - 2n] \rfloor$  QSC, where  $(a_l, a_r) < 2n$ .  $\square$

**Example 1** Let  $q = 3$ ,  $n = 13$ , then

$$x^{13} + 1 = (x + 1)Q_{26}(x) = (x + 1)m_1^{(26)}(x)m_5^{(26)}(x)m_7^{(26)}(x)m_{17}^{(26)}(x).$$

Since  $-1 \notin C_1^{(26)}$ , then

$$Q_{26}(x) = m_1^{(26)}(x)m_1^{(26)*}(x)m_5^{(26)}(x)m_5^{(26)*}(x),$$

where  $m_1^{(26)*}(x) = m_{17}^{(26)}(x)$ ,  $m_5^{(26)*}(x) = m_7^{(26)}(x)$ ,

$$i.e., x^{13} + 1 = (x + 1)m_1^{(26)}(x)m_1^{(26)*}(x)m_5^{(26)}(x)m_5^{(26)*}(x).$$

Let  $g_2(x) = m_1^{(26)}(x)m_5^{(26)}(x)$  and  $g_4(x) = m_1^{(26)}(x)$ . Then  $C_2 = \langle g_2(x) \rangle$  is a ternary [4, 7, 13] negacyclic code and  $C_4 = \langle g_4(x) \rangle$  is a ternary [3, 10, 13] negacyclic code, and  $C_2^\perp \subset C_2 \subset C_4$ . And

$$x^{13} - 1 = (x - 1)m_2^{(26)}(x)m_4^{(26)}(x)m_8^{(26)}(x)m_{14}^{(26)}(x),$$

where  $m_2^{(26)*}(x) = m_8^{(26)}(x)$  and  $m_4^{(26)*}(x) = m_{14}^{(26)}(x)$ .

Then

$$x^{13} - 1 = (x - 1)m_2^{(26)}(x)m_4^{(26)}(x)m_2^{(26)*}(x)m_4^{(26)*}(x).$$

Let  $g_1(x) = m_2^{(26)}(x)m_4^{(26)}(x)$  and  $g_3(x) = m_2^{(26)}(x)$ . Then  $C_1 = \langle g_1(x) \rangle$  is a ternary [4, 7, 13] cyclic code,  $C_3 = \langle g_3(x) \rangle$  is a ternary [3, 10, 13] cyclic code, and  $C_1^\perp \subset C_1 \subset C_3$ . Define  $f(x) = m_4^{(26)}(x)m_5^{(26)}(x)$  and  $(C_1 \vee C_2)^\perp \subset C_1 \vee C_2 \subset C_3 \vee C_4$ . We can obtain that  $C_1 \vee C_2$  is a [26, 14, 4] cyclic code and  $C_3 \vee C_4$  is a [26, 20, 3] cyclic code. By Theorem 3.6 and 3.7 we have an  $(a_p, a_r) - [[26 + a_l + a_r, 2]]_7$  QSC, and  $\text{ord}(f(x)) = 2n$ . The synchronization capabilities of  $(a_p, a_r) - [[26 + a_l + a_r, 2]]_7$  can reach the upper bound. And that can correct up to 1 bit error and 1 phase error

**Example 2** Let  $q = 37$ ,  $n = 41$ , then

$$x^{41} + 1 = (x + 1)Q_{82}(x) = (x + 1)m_1^{(82)}(x)m_3^{(82)}(x)m_5^{(82)}(x)m_{11}^{(82)}(x)m_{15}^{(82)}(x)m_{17}^{(82)}(x)m_{23}^{(82)}(x)m_{33}^{(82)}(x).$$

Since  $-1 \notin C_1^{(26)}$ , then

$$Q_{82}(x) = m_1^{(82)}(x)m_3^{(82)}(x)m_5^{(82)}(x)m_{15}^{(82)}(x)m_1^{(82)*}(x)m_3^{(82)*}(x)m_5^{(82)*}(x)m_{15}^{(82)*}(x),$$

Let  $g_2(x) = m_1^{(82)}(x)m_3^{(82)}(x)$  and  $g_4(x) = m_1^{(82)}(x)$ . Then  $C_2 = \langle g_2(x) \rangle$  is a [41, 31, 6] negacyclic code over  $F_{37}$ ,  $C_4 = \langle g_4(x) \rangle$  is a [41, 36, 4] negacyclic code over  $F_{37}$ , and  $C_2^\perp \subset C_2 \subset C_4$ . And

$$x^{41} - 1 = (x - 1)m_2^{(82)}(x)m_4^{(82)}(x)m_6^{(82)}(x)m_8^{(82)}(x)m_{10}^{(82)}(x)m_{12}^{(82)}(x)m_{22}^{(82)}(x)m_{30}^{(82)}(x).$$

where  $m_2^{(82)*}(x) = m_8^{(82)}(x)$ ,  $m_4^{(82)*}(x) = m_{10}^{(82)}(x)$ ,  $m_6^{(82)*}(x) = m_{22}^{(82)}(x)$  and  $m_{12}^{(82)*}(x) = m_{30}^{(82)}(x)$ .

Then

$$x^{41} - 1 = (x - 1)m_2^{(82)}(x)m_4^{(82)}(x)m_6^{(82)}(x)m_{12}^{(82)}(x)m_2^{(82)*}(x)m_4^{(82)*}(x)m_6^{(82)*}(x)m_{12}^{(82)*}(x).$$

Let  $g_1(x) = m_2^{(82)}(x)m_4^{(82)}(x)$  and  $g_3(x) = m_2^{(82)}(x)$ . Then  $C_1 = \langle g_1(x) \rangle$  is a [41, 31, 6] cyclic code over  $F_{37}$ ,  $C_3 = \langle g_3(x) \rangle$  is a [41, 36, 4] cyclic code over  $F_{37}$ , and  $C_1^\perp \subset C_1 \subset C_3$ . Defined  $f(x) = m_3^{(82)}(x)m_4^{(82)}(x)$  and  $(C_1 \vee C_2)^\perp \subset C_1 \vee C_2 \subset C_3 \vee C_4$ . We can obtain that  $C_1 \vee C_2$  is a [82, 62, 6] cyclic code, and  $C_3 \vee C_4$  is a [82, 72, 4] cyclic code. By Theorem 3.6 and 3.7 we have an  $(a_p, a_r) - [[82 + a_l + a_r, 42]]_{37}$  QSC, and  $\text{ord}(f(x)) = 2n$ . The synchronization capabilities of  $(a_p, a_r) - [[82 + a_l + a_r, 42]]_{37}$  can reach the upper bound. And that can correct up to 1 bit error and 2 phase errors.

## 4 The Second Class of QSCs

Let  $q$  be an odd prime power and  $(n, q) = 1$ . Let  $\xi$  be an  $2n$ -th primitive root of unity. A negacyclic BCH code of length  $n$  over  $F_q$  with designed distance  $\delta$  is a negacyclic code with generator polynomial  $g(x) = \prod_{j \in Z_1} (x - \xi^j)$ ,  $Z_1 = \bigcup_{j=b}^{b+\delta-1} C_{1+2j}^{2n}$ . And a BCH code of length  $n$  over  $F_q$  with designed distance  $\delta$  is a cyclic code with generator polynomial  $g^*(x) = \prod_{j \in Z_2} (x - \xi^j)$ ,  $Z_2 = \bigcup_{j=b}^{b+\delta-1} C_{2j}^{2n}$ .

In this section we assume  $n = \frac{q^{2m}-1}{q-1}$ .

**Lemma 4.1** [14] Let  $q$  be an odd prime power. Let  $n = \frac{q^{2m}-1}{q-1}$  with  $m \geq 2$ . If  $C$  be a  $q$ -ary BCH code of length  $n$  with designed distance  $\delta$ , then  $C^\perp \subset C$  if and only if  $2 \leq \delta \leq \delta_{\max} = \frac{q^m-1}{q-1}$ .

According to the above lemma, we can construct a cyclic BCH code.

**Theorem 4.2** Let  $C_1 = \langle g_1(x) \rangle$  and  $C_3 = \langle g_3(x) \rangle$  be cyclic codes of length  $n = \frac{q^{2m}-1}{q-1}$  over  $F_q$ , where

$$g_1(x) = \text{lcm}(m_0(x), m_2(x), \dots, m_{2(\delta-2)}(x))$$

$$g_3(x) = \text{lcm}(m_0(x), m_2(x), \dots, m_{2(\delta-i-2)}(x)),$$

$$1 \leq i \leq \delta - 2$$

If  $2 \leq \delta \leq \delta_{\max} = \frac{q^m-1}{q-1}$ , then  $C_1$  is a  $\left[ \frac{q^{2m}-1}{q-1}, \frac{q^{2m}-1}{q-1} - 2m \left[ (\delta-1)(1-q^{-1}) \right], d_1 \geq \delta \right]$  cyclic code and  $C_3$  is a  $\left[ \frac{q^{2m}-1}{q-1}, \frac{q^{2m}-1}{q-1} - 2m \left[ (\delta-i-1)(1-q^{-1}) \right], d_3 \geq \delta-i \right]$  cyclic code. Moreover  $C_1 \perp \subset C_1 \subset C_3$ .

**Proof** It's obvious that  $g_1(x) = \text{lcm}(m_0(x), m_2(x), \dots, m_{2(\delta-2)}(x))$  and  $g_3(x) = \text{lcm}(m_0(x), m_2(x), \dots, m_{2(\delta-i-2)}(x))$ , then  $g_3(x) | g_1(x)$ . Hence  $C_1 \subset C_3$ . By the lemma 4.1,  $C_1^\perp \subset C_1$ . Thus,  $C_1^\perp \subset C_1 \subset C_3$ .  $\square$

**Lemma 4.3** [19] Let  $n = \frac{q^{2m}-1}{q-1}$ , and let  $q$  be an odd prime power, where  $m \geq 1$ . Let  $C$  be a negacyclic BCH code of length  $n$  with designed distance  $\delta^*$ , where  $2 \leq \delta^* \leq \delta_{\max}^* = \frac{q^{m+1}+q^m+q-3}{2(q-1)}$ . Let  $\gamma = \left\lfloor \frac{(\delta^*-2)(q-1)}{q^m-1} \right\rfloor$ . Then  $C$  be a negacyclic BCH code, and has dimension

$$k = \begin{cases} n - 2m \left\lfloor \left( \delta^* - \frac{3}{2} \right) (1 - q^{-1}) \right\rfloor & \gamma < \left\lfloor \frac{q-1}{4} \right\rfloor \\ n - 2m \left\lfloor \left( \delta^* - \frac{3}{2} \right) (1 - q^{-1}) \right\rfloor + \frac{m}{2} (4\gamma - q + 3) & \left\lfloor \frac{q-1}{4} \right\rfloor \leq \gamma \leq \frac{q-3}{2} \\ n - 2m \left\lfloor \left( \delta^* - \frac{3}{2} \right) (1 - q^{-1}) \right\rfloor + \frac{m}{2} (q - 3) & \gamma = \frac{q-1}{2} \end{cases}.$$

**Lemma 4.4** [19] Let  $n = \frac{q^{2m}-1}{q-1}$ , where  $m \geq 2$ . Let  $C$  be a  $q$ -ary negacyclic BCH code with designed distance  $\delta^*$ , then  $C^\perp \subset C$  if and only if  $2 \leq \delta^* \leq \delta^*_{\max} = \frac{q^{m+1}+q^m+q-3}{2(q-1)}$ .

According to the above lemmas, we can construct a negacyclic dual-containing BCH code.

**Theorem 4.5** Let  $C_2 = \langle g_2(x) \rangle$  and  $C_4 = \langle g_4(x) \rangle$  be negacyclic codes of length  $n = \frac{q^{2m}-1}{q-1}$  over  $F_q$ , where

$$g_2(x) = lcm(m_1(x), m_3(x), \dots, m_{1+2(\delta^*-2)}(x))$$

$$g_4(x) = lcm(m_1(x), m_3(x), \dots, m_{1+2(\delta^*-j-2)}(x))$$

$$2 \leq j \leq \delta^* - 2,$$

If  $2 \leq \delta^* \leq \delta^*_{\max} = \frac{q^{m+1}+q^m+q-3}{2(q-1)}$ , then  $C_2$  is a  $[\frac{q^{2m}-1}{q-1}, k_2, d_2 \geq \delta^*]$  negacyclic code and  $C_4$  is a  $[\frac{q^{2m}-1}{q-1}, k_4, d_4 \geq \delta^* - i]$  negacyclic code. Moreover  $C_2^\perp \subset C_2 \subset C_4$ .

**Proof** Since  $g_2(x) = lcm(m_1(x), m_3(x), \dots, m_{1+2(\delta^*-2)}(x))$  and  $g_4(x) = lcm(m_1(x), m_3(x), \dots, m_{1+2(\delta^*-j-2)}(x))$ , then  $g_4(x) | g_2(x)$ , i.e.,  $C_2 \subset C_4$ , by Lemma 4.4, we can obtain  $C_2^\perp \subset C_2 \subset C_4$ .  $\square$

Let the notations be as above Theorems.  $C_1 \vee C_2$  is a  $[2\frac{q^{2m}-1}{q-1}, k_1 + k_2, d]$  cyclic code with  $d = \min\{2d_1, 2d_2, \max\{d_1, d_2\}\}$ , and  $C_3 \vee C_4$  is a  $[2\frac{q^{2m}-1}{q-1}, k_3 + k_4, d^*]$  cyclic code with  $d^* = \min\{2d_3, 2d_4, \max\{d_3, d_4\}\}$ , where  $k_1 = \frac{q^{2m}-1}{q-1} - 2m[(\delta-1)(1-q^{-1})]$ ,  $k_3 = \frac{q^{2m}-1}{q-1} - 2m[(\delta-i-1)(1-q^{-1})]$  and the value of  $k_2, k_4$  can be obtained by the value of  $\gamma = \left\lfloor \frac{(\delta^*-2)(q-1)}{q^m-1} \right\rfloor$ , then  $C_1^\perp \vee C_2^\perp \subset C_1 \vee C_2 \subset C_3 \vee C_4$ .

**Theorem 4.6** There exists an  $(a_l, a_r) - [[2\frac{q^{2m}-1}{q-1} + a_l + a_r, 2(k_1 + k_2) - 2\frac{q^{2m}-1}{q-1}]]$  QSC that can correct up to  $\left\lfloor \frac{d^*-1}{2} \right\rfloor$  bit errors and  $\left\lfloor \frac{d-1}{2} \right\rfloor$  phase errors, where  $(a_l + a_r) < 2n$ .  $d = \min\{2d_1, 2d_2, \max\{d_1, d_2\}\}$ ,  $d^* = \min\{2d_3, 2d_4, \max\{d_3, d_4\}\}$ .

**Example 3** Let  $q = 3$ ,  $m = 2$ , and  $n = \frac{q^{2m}-1}{q-1} = 40$ .

Then  $\delta_{\max} = \frac{q^m-1}{q-1} = 4$ . We suppose  $\delta = 4$ ,  $i = 1$ , then

$$g_1(x) = lcm(m_0^{(80)}(x), m_2^{(80)}(x), m_4^{(80)}(x)), g_3(x) = lcm(m_0^{(80)}(x), m_2^{(80)}(x)),$$

and  $g_3(x) | g_1(x)$ . It's clear that  $C_1 = \langle g_1(x) \rangle$  is a  $[40, 32, 4]$  BCH code over  $F_3$ ,  $C_3 = \langle g_3(x) \rangle$  is a  $[40, 36, 3]$  BCH code over  $F_3$ , and  $C_1^\perp \subset C_1 \subset C_3$ .

By  $2 \leq \delta^* \leq \delta^*_{\max} = \frac{q^{m+1}+q^m+q-3}{2(q-1)}$ , we suppose  $\delta^* = 4$ ,  $j = 1$ , then

$$g_2(x) = lcm(m_1^{(80)}(x), m_3^{(80)}(x), m_5^{(80)}(x)), g_4(x) = lcm(m_1^{(80)}(x), m_3^{(80)}(x)),$$

and  $g_4(x) | g_2(x)$ . It's clear that  $C_2 = \langle g_2(x) \rangle$  is a  $[40, 32, 4]$  negacyclic BCH code over  $F_3$ ,  $C_4 = \langle g_4(x) \rangle$  is a  $[40, 36, 3]$  negacyclic BCH code over  $F_3$ , and  $C_2^\perp \subset C_2 \subset C_4$ .

Defined  $f(x) = m_4^{(80)}(x)m_5^{(80)}(x)$  and  $(C_1 \vee C_2)^\perp \subset C_1 \vee C_2 \subset C_3 \vee C_4$ . We can obtain that  $C_1 \vee C_2$  is a  $[80, 64, 4]$  cyclic code, and  $C_3 \vee C_4$  is a  $[80, 72, 3]$  cyclic code. By Theorem 3.6 and 3.7 we have an  $(a_l, a_r) - [[80 + a_l + a_r, 48]]_3$  QSC, and  $\text{ord}(f(x)) = 2n$ . The synchronization capabilities of  $(a_l, a_r) - [[80 + a_l + a_r, 48]]_3$  can reach the upper bound. And that can correct up to 1 bit error and 1 phase errors.

## 5 Conclusion

In this paper, we obtained two classes of QSCs based on  $(u + v)|(u - v)$  construction by using cyclic codes and negacyclic codes. We obtain a method to find dual-containing cyclic and negacyclic codes of length odd prime and get a class of QSCs. Besides, we use BCH codes and negacyclic BCH codes of length  $\frac{q^{2m}-1}{q-1}$  to construct QSCs. And their synchronization error tolerance can achieve the best attainable misalignment tolerance under certain constraints.

**Acknowledgements** This work is supported by the National Natural Science Foundation of China under Grant Nos U21A20428, 12171134.

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